Data Visualization — Final Project

📋 PART 1: General Requirements

1.1 Project Objective

Each team is required to build a **fully interactive Plotly Dash dashboard** that applies all data visualization concepts covered throughout the course (Week 1 – Week 9) on a **single real-world dataset**. The dashboard must demonstrate the team's ability to choose the right chart type for the right analytical purpose, create meaningful visualizations, and extract insights from data.

1.2 Course Topics Coverage

Your dashboard **must include all** of the following chart types covered in the course. Each chart must be applied meaningfully to your dataset with a clear analytical purpose.

⚠️ ***Important :*** You Should Implement **the Guidelines** for each chart which covered in **Lectures slides.**

Week	**Topic**	**Required Chart Type(s)**
Week 1	Comparison Charts	Column Chart **and** Bar Chart
Week 2	Comparison Charts	Stacked Column Chart, Stacked Bar Chart, Clustered Column Chart, **and** Clustered Bar Chart
Week 3	Relationship Charts	Scatter Chart
Week 4	Relationship Charts	Bubble Chart
Week 5	Distribution Charts	Histogram Chart
Week 6	Distribution Charts	Box Chart
Week 7	Distribution Charts	Violin Chart

Week 8	Time-Series Charts	Line Chart
Week 9	Time-Series Charts	Area Chart

⚠️ ***Important:*** *Every single chart type listed above must appear in your final dashboard. Missing any chart type will result in mark deduction.*

1.3 Dashboard Requirements

Your Plotly Dash application must meet the following criteria:

Functional Requirements

- All charts must be created using **Plotly** (Plotly Express or Plotly Graph Objects)
- The dashboard must be built using **Dash by Plotly**
- The dashboard must include **at least 3 interactive elements** (dropdowns, sliders, radio buttons, checklists, etc.)
- Interactive elements must be connected to charts through **Dash callbacks** so that charts update dynamically based on user input
- Each chart must have a **clear and descriptive title**
- All axes must be **properly labeled**
- Charts must include **appropriate legends** where applicable
- Charts must use **appropriate color schemes** that are readable and consistent

Layout Requirements

- The dashboard must have a **clear header** with the project title
- Charts must be **logically organized** (not randomly placed)
- The layout must be **clean, readable, and visually appealing**
- The dashboard must be **responsive** and viewable without horizontal scrolling

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1.4 Deliverables

#	Deliverable	Format	Description
1	**Source Code**	.py files	Complete Dash application code, well-organized and commented
2	**Data Preprocessing Notebook**	.ipynb	Jupyter Notebook showing all data cleaning, transformation, and preparation steps
3	**Final Dataset**	.csv or .xlsx	The cleaned and processed dataset used in the dashboard
4	**Documentation**	README.md or Word	Project overview, dataset description, team member roles, and instructions to run the dashboard

Each team must submit the following:

1.5 Submission Guidelines

- Inside the folder structure should be:

```text
Team03_COVID19Analysis/
├── app.py                  # Main Dash application
├── data/
│   ├── raw_data.csv        # Original dataset
│   └── cleaned_data.csv    # Processed dataset
├── notebooks/
│   └── preprocessing.ipynb # Data cleaning notebook
├── assets/
│   └── style.css           # (Optional) Custom CSS
├── README.md               # Documentation
├── requirements.txt        # Python dependencies
```

1.6 Academic Integrity

- All work must be **original** and created by the team members
- Proper **citation** is required for any external code, tutorials, or resources used
- **Plagiarism** or copying from other teams will result in **zero marks** for all involved parties
- Using AI tools for code assistance is permitted, but you must **understand and be able to explain** every line of your code during the Q&A session

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1.7 Technical Setup

Running the Dashboard

```text
python app.py
```

The dashboard will be accessible at `http://127.0.0.1:8050/`

PART 2: Project Descriptions

*Each team will be assigned **one** of the following projects. Each project specifies a dataset and the analytical domain. Your job is to apply **all required chart types** (listed in Section 1.2) meaningfully to the given dataset and build an interactive Plotly Dash dashboard.*

🔴 Project 1: Global COVID-19 Pandemic Analysis

Description

This project focuses on analyzing the global spread, impact, and recovery patterns of the COVID-19 pandemic. Teams will explore confirmed cases, deaths, vaccinations, and testing data across countries and continents. The dashboard should allow users to compare pandemic metrics between countries, investigate the relationship between socio-economic factors (GDP, population density) and pandemic outcomes, examine the distribution of key health metrics across regions, and track the progression of the pandemic over time. The goal is to build a comprehensive visual analytics tool that tells the story of the pandemic through data.

Dataset

- **Name:** Our World in Data — COVID-19 Dataset
- **Source:** https://github.com/owid/covid-19-data/tree/master/public/data
- **File:** owid-covid-data.csv

- **Size:** ~200,000+ rows | 67 columns

🟠 Project 2: World Happiness Report Analysis

Description

This project explores the factors that contribute to national happiness across the globe. Teams will analyze data from the World Happiness Report to understand how economic output, social support, personal freedom, life expectancy, generosity, and corruption perception influence a country's happiness score. The dashboard should enable users to compare happiness rankings across countries and regions, investigate which factors are most strongly associated with happiness, explore how happiness metrics are distributed across the world, and observe how happiness trends have changed over multiple years. The goal is to uncover what makes nations happy through visual exploration.

Dataset

- **Name:** World Happiness Report (2015–2023)
- **Source:** https://www.kaggle.com/datasets/unsdsn/world-happiness
- **Files:** Multiple CSV files (one per year), to be merged

🟡 Project 3: E-Commerce Sales Performance Dashboard

Description

This project analyzes the sales performance of a retail superstore operating across multiple regions. Teams will explore sales, profits, discounts, product categories, customer segments, and shipping modes to uncover business insights. The dashboard should allow users to compare performance across product categories and regions, investigate the relationship between discounting strategies and profitability, analyze the distribution of order values and profit margins, and track sales and profit trends over time. The goal is to build an analytical dashboard that could help business managers make data-driven decisions.

Dataset

- **Name:** Sample Superstore Dataset
- **Source:** https://www.kaggle.com/datasets/vivek468/superstore-dataset-final

- **File:** Sample - Superstore.csv
- **Size:** ~10,000 rows | 21 columns

🟢 Project 4: Global Education Statistics Dashboard

Description

This project explores education indicators across countries worldwide, including literacy rates, school enrollment, government spending on education, and student-teacher ratios. Teams will investigate how education metrics vary across countries and continents, examine the relationship between economic development and educational outcomes, analyze the distribution of key education indicators, and track how global education has improved over decades. The goal is to provide a visual understanding of the global state of education and identify areas that need attention.

Dataset

- **Name:** World Education Data
- **Source:** https://www.kaggle.com/datasets/nelgiriyewithana/world-educational-data
- **File:** Global_Education.csv
- **Size:** 202 rows | 29 columns

🔵 Project 5: Spotify Music Trends & Audio Features Analysis

Description

This project dives into the world of music by analyzing Spotify track data, including audio features such as danceability, energy, tempo, and valence, as well as popularity metrics and genre classifications. Teams will compare popularity and audio characteristics across artists and genres, explore the relationships between different audio features and a song's popularity, examine how audio features are distributed across the music library, and track how music trends and preferences have evolved over the years. The goal is to uncover patterns in what makes music popular and how musical tastes have shifted over time.

Dataset

- **Name:** Spotify Tracks Dataset

- **Source:** https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset
- **Size:** ~100,000+ tracks

🟣 Project 6: Global Health & Life Expectancy Dashboard

Description

This project examines health indicators across countries, focusing on life expectancy and the factors that influence it, such as mortality rates, disease prevalence, economic conditions, and lifestyle factors like alcohol consumption and BMI. Teams will compare health outcomes across developed and developing nations, investigate how socio-economic factors relate to life expectancy, analyze the distribution of health metrics across different populations, and track how global health has improved or declined over time. The goal is to identify the key determinants of a healthy population through visual data analysis.

Dataset

- **Name:** Life Expectancy (WHO) Dataset
- **Source:** https://www.kaggle.com/datasets/lashagoch/life-expectancy-who-updated
- **Size:** ~3000 rows | 21 columns

🟤 Project 7: Air Quality & Environmental Pollution Dashboard

Description

This project analyzes global air quality data across cities and countries, examining pollutant levels including PM2.5, ozone, nitrogen dioxide, and carbon monoxide. Teams will compare air quality indices across the most and least polluted cities in the world, explore the relationship between urbanization factors and pollution levels, analyze the distribution of various pollutant concentrations, and track how air quality changes over time and across seasons. The goal is to create an environmental awareness tool that visually communicates the state of global air pollution.

Dataset

- **Name:** Global Air Pollution Dataset

- **Source:** https://www.kaggle.com/datasets/hasibalmuzdadid/global-air-pollution-dataset
- **Size:** ~23,000 rows | 12 columns

⚫ Project 8: FIFA Players Statistics Dashboard

Description

This project analyzes football (soccer) player statistics from the FIFA video game series, covering player ratings, physical attributes, skill metrics, market values, and club affiliations. Teams will compare player ratings and attributes across clubs, leagues, and nationalities, investigate the relationship between player age, skill attributes, and market value, analyze how player ratings and physical characteristics are distributed across different positions, and track how player ratings evolve across different FIFA editions if multiple years are used. The goal is to provide a scouting-like analytics dashboard for exploring player data.

Dataset

- **Name:** FIFA 23 Complete Player Dataset
- **Source:** https://www.kaggle.com/datasets/stefanoleone992/fifa-23-complete-player-dataset
- **File:** male_players.csv
- **Size:** ~18,000 players

🔶 Project 9: Global Energy Production & Consumption Dashboard

Description

This project explores how countries around the world produce and consume energy from various sources including fossil fuels (oil, gas, coal), nuclear power, and renewable sources (solar, wind, hydro). Teams will compare energy consumption and renewable adoption rates across countries, investigate the relationship between economic development and energy use patterns, analyze the distribution of energy metrics like per capita consumption and carbon intensity, and track the global energy transition from fossil fuels to renewables over decades. The goal is to build a dashboard that visualizes the world's energy landscape and the progress toward clean energy.

Dataset

- **Name:** Our World in Data — Energy Dataset
- **Source:** https://github.com/owid/energy-data or https://www.kaggle.com/datasets/pralabhpoudel/world-energy-consumption
- **File:** World Energy Consumption.csv
- **Size:** ~16,000 rows | 120+ columns

🔷 Project 10: Movie Industry Analysis Dashboard

Description

This project analyzes the movie industry by exploring movie ratings, revenues, budgets, genres, runtimes, and production details from a large database of films. Teams will compare financial performance and ratings across genres, production companies, and languages, investigate the relationship between movie budgets and their box office returns, analyze how ratings, runtimes, and revenues are distributed across the film industry, and track how the movie industry has evolved over the years in terms of output, revenue, and audience preferences. The goal is to create an entertainment industry analytics dashboard that reveals what makes movies successful.

Dataset

- **Name:** TMDB 5000 Movie Dataset
- **Source:** https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata
- **Size:** ~5,000 movies